Balancing Customer Engagement and Annoyance in Online Retail: Insights from a Field Experiment

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The emergence of e-commerce and digital operations has given firms unprecedented reach: retailers can now send real-time emails and mobile notifications to customers anywhere in the world. These direct-to-consumer promotions (e.g., coupons, emails, push alerts) can boost engagement and sales, but when overused, they can create annoyance and notification fatigue, alienate customers, increase unsubscriptions, and ultimately weaken long-term customer spending. We first conduct a large-scale randomized field experiment at an online retailer to study this trade-off. We find that sending fewer emails—once every two days instead of every day—reduces unsubscription by 59%, but at the cost of a 5–8% decrease in short-term revenue and purchase rate. These effects are heterogeneous: reducing coupon-email frequency lowers unsubscription, especially among newer customers and those who open emails, and it reduces short-run spending, particularly among recent purchasers. Then, using historical data over a longer time frame, we find that unsubscribing is followed by a large decline in monthly revenue, equivalent to 41% of average monthly spending. We then embed these results in a customer lifetime value (CLV) optimization model. The model shows that personalizing email frequency using the field-experimental estimates can improve long-term value: a myopic policy (maximizing immediate revenue) raises average CLV by 6.8%, while a full CLV-optimal policy yields an 8.3% gain over the daily-email baseline. Overall, our findings highlight the importance of balancing short-term revenue with customer retention: sending emails too frequently can boost immediate sales but may "burn out" customers, whereas moderately reducing email frequency can yield higher lifetime value.

Key words: field experiment, customer annoyance, email frequency, customer lifetime value, online retail operations.

1. Introduction

As retail has digitized and e-commerce has expanded, firms can communicate with customers instantly and frequently through email, mobile apps, and social media. Many retailers now rely on automated newsletters, push notifications, and triggered messages (e.g., cart reminders, price-drop alerts, personalized coupons/emails) to bring shoppers back to the platform. These digital nudges can stimulate repeat purchases and bolster short-run sales because the marginal cost of sending an additional message is low and experimentation is easy. Yet the same scalability creates a fundamental tension: when promotions arrive too often or feel poorly targeted, customers experience annoyance and message fatigue, begin to ignore communications, unsubscribe from email lists, disable notifications, or delete the app altogether. A prominent illustration comes from Groupon. In an interview reflecting on the company's growth, cofounder Andrew Mason described internal pressure to increase deal and email frequency, recalling his initial reaction to the idea of sending two emails per day as "That sounds awful. Who wants to get two emails every single day from a company?" (Blumberg 2018). He nevertheless agreed to run controlled tests, and the short-run results looked favorable: unsubscribes rose only slightly while incremental purchases more than offset the immediate attrition, making higher frequency appear "rational" by near-term metrics (Blumberg 2018). Mason's retrospective takeaway, however, was that hypergrowth left little time to observe longer-run churn dynamics, so the cumulative erosion of goodwill surfaced later, after repeated exposure had already trained customers to tune out or opt out; in his words, the service eventually became a "vestige of what it once was" (Blumberg 2018). The Groupon episode highlights why optimizing promotional intensity based solely on short-run lift can be misleading when annoyance and fatigue compound over time.

Similar tensions have surfaced well beyond Groupon. In the daily-deals ecosystem, the ease of "broadcasting" promotions at scale contributed to consumer annoyance and inbox saturation, with reported email response rates falling as customers became desensitized to constant deal blasts (Schonfeld 2011). Related e-commerce models underscore how deeply frequent outreach can become embedded in operating strategy: flash-sale retailer Zulily reportedly experienced revenue headwinds when deliverability problems reduced its ability to send daily marketing emails, illustrating both the power of high-frequency communication for short-run sales and the fragility of a model that relies on constant contact (Cook 2014). Firms have treated message volume itself as a design lever to manage annoyance and protect long-run engagement. Fab, the design e-commerce company, was reported to proactively reduce emails to users who were not opening them, and LinkedIn described cutting a substantial share of its outgoing emails in response to member complaints, reflecting an emerging managerial consensus that the incremental lift from additional messages must be weighed against the cumulative cost of fatigue and opt-outs (Weber 2013, Awan 2015).

This context, in which ubiquitous digital promotion is accompanied by the risk of overwhelming customers, motivates our study. We examine the engagement–annoyance trade-off in online retail by leveraging both a randomized experiment and observational data from a retailer.[1] In a field experiment, we randomly assigned 603,153 subscribed customers to receive either the status-quo daily coupon email or a reduced every-other-day schedule for a week. We measure how this frequency decision affects short-term outcomes (daily revenue and purchase rates) and customers' rate of unsubscribing from the email list. Then, using the retailer's historical data, we quantify the long-term consequences of unsubscribing for customers' future purchases and retention. We integrate these findings into a customer lifetime value (CLV) optimization framework. Using the estimated effects of email frequency on short-term revenue and unsubscription (from the experiment) and unsubscription's effects on revenue and retention (from the historical analysis), we compare four email policies: (1) the baseline (daily emails to all), (2) a uniform every-other-day schedule for all, (3) a myopic policy that personalizes email frequency to maximize immediate expected revenue, and (4) a full CLV-optimal policy that personalizes frequency to maximize expected lifetime revenue accounting for retention. We evaluate the two personalized policies both with and without a cap on coupon redemptions.

Our main findings are threefold. First, the field experiment confirms a clear trade-off: sending emails daily rather than every other day increases purchase probability and daily sales but raises customers' probability of unsubscribing (i.e., opting out of the email list). The more aggressive promotional policy drives short-term revenue at the cost of losing some customers. The every-other-day schedule lowers the daily opt-out rate by 59% and lowers purchase incidence and revenue during the week by 5–8%. These effects are heterogeneous, and they fall on different customers. Reducing the frequency of coupon emails mitigates unsubscribing most among newer customers and among customers who open the emails, whereas it lowers short-run spending among customers who had purchased in the pre-experiment week, for whom the opt-out treatment effect does not differ significantly from that for other customers. Second, our analysis of historical panel data shows that customers who unsubscribe from promotional emails spend much less thereafter. Using regression, instrumental-variable, and within-customer methods, we find that unsubscribing is followed by a decline in monthly revenue of \$9.31 (our instrumental-variable estimate), equivalent to 41% of the panel-wide monthly mean, and by a fall of more than half in the probability of repurchasing within the year. Third, by embedding these estimates in a CLV optimization framework, we show that an optimized, personalized email policy can improve the firm's CLV. In an unconstrained setting (no budget limit on coupon redemptions), the myopic policy boosts average CLV by 6.8% over the

[1] Under our non-disclosure agreement, we cannot disclose the retailer's name; it is a major North American e-commerce retailer.

baseline, while the full CLV-optimal policy yields an 8.3% improvement. With coupon redemptions held at status-quo levels, the gains are smaller but still positive (2.3% and 4.5%). Nearly all of the unconstrained gain is captured by a simple rule that keeps daily emails for recently active customers and moves everyone else to every other day. We also conduct a sensitivity analysis: as the baseline customer retention rate increases (i.e., when customers are more loyal), the relative advantage of reducing email frequency grows. When retention is high, protecting it with less aggressive emailing yields larger lifetime revenue gains.

Overall, our contributions are both empirical and methodological. Empirically, to our knowledge, we provide the first large-scale randomized field experiment in a retail setting that identifies how email frequency affects short-run revenue and directly observed opt-out behavior, documents heterogeneity in these effects across customer types, and links these experimental estimates to historical evidence on the longer-run revenue and retention consequences of opting out. Methodologically, we show how to combine field-experimental and observational data in an optimization of CLV, highlighting how direct causal estimates of engagement and unsubscription can inform customer relationship management decisions, and we state and test the assumptions that this combination requires.

It is worth noting that a distinctive feature of our setting is that we observe a direct and explicit signal of customer annoyance through email unsubscriptions. This allows us to account for annoyance not only through subsequent purchasing behavior, which is a much more indirect proxy, but also through a clear behavioral action that reflects disengagement. Much of the existing literature must instead infer annoyance implicitly: Todri et al. (2020) find that overly frequent online ads can reduce downstream purchases, but they model annoyance as a latent state in a hidden Markov model, which introduces additional identification challenges. Zhang et al. (2017) likewise infer the effect of email frequency on customer states from observational data. However, in this paper, unsubscribing is an observable behavioral outcome of annoyance, and we can estimate what it costs the retailer.

2. Related Literature and Positioning

As discussed above, frequent marketing contact can encourage purchases but may also prompt customers to disengage. With this trade-off in mind, we organize the literature review around two central questions: how contact frequency affects customer behavior, and how firms should incorporate these effects when designing contact policies. Table 1 positions our study relative to the closest related work. We then discuss the broader literature on promotion decisions and customer lifetime value.

Table 1 Closest Related Studies

Study	Lever	Identification	Disengagement measure	Outcome
Godfrey et al. (2011)	Contact frequency	Observational	Inferred (ideal point)	Repurchase
Zhang et al. (2017)	Email frequency	Observational (HMM)	Latent state	Profitability
Todri et al. (2020)	Ad exposures	Observational (HMM)	Latent state	Purchase funnel
Sahni et al. (2019)	Ad frequency	Experimental	None (no opt-out)	Site visits
Sahni et al. (2017)	Offer (yes/no)	Experimental	None	Purchases
Wang et al. (2022)	Email engagement	Experimental	None	Profitability
Simester et al. (2006)	Catalog mailing	Model + field test	None	Long-run profit
This paper	**Email frequency**	**Experiment; historical models**	**Observed opt-out**	**Revenue, retention, CLV**

"Identification" is the source of variation in the lever; HMM denotes a hidden Markov model. "Disengagement measure" is how the study captures a customer's negative reaction to contact.

2.1. Contact Frequency and Customer Disengagement

Research shows that the returns to additional marketing contact need not remain positive. Godfrey et al. (2011) identify an inverted-U relationship between multichannel communication and repurchasing. Zhang et al. (2017) model email frequency, customer states, and profitability, while Todri et al. (2020) examine the benefits and annoyance associated with repeated display-ad exposure. These studies show why contact decisions should account for changing customer responsiveness. The behavioral basis for a negative frequency response is well documented: repeated exposure produces wearout (Pechmann and Stewart 1988), unsolicited contact imposes attention costs (Van Zandt 2004), and intrusive messages trigger psychological reactance (Brehm 1966, Edwards et al. 2002), which can lead customers to withdraw from the relationship. van Diepen et al. (2009) find that a charity's direct mailings raise donations in the short run but cannibalize the revenue of its subsequent mailings, so the return to another mailing is smaller than its immediate response suggests. More broadly, Ashlagi et al. (2020) study how match recommendations and informative signaling reduce communication congestion in matching markets.[2]

The closest experimental studies vary related levers but cannot observe opt-out. Sahni et al. (2019) randomize the frequency and timing of retargeted display ads, a channel in which the viewer has no opt-out to exercise. Sahni et al. (2017) show that emailed discount offers act partly as advertising, the mechanism we invoke for the revenue effect of a reminder, and Sahni et al. (2018) test personalization of email content. Wang et al. (2022) experimentally vary email engagement in a subscription-service setting and study its effects on customer retention, service consumption, and profitability. Zhang et al. (2017) come closest in question, modeling how email frequency moves customers between latent responsiveness states and the resulting profit implications, but their disengagement state is inferred from observational purchase data, as are the annoyance thresholds

[2] Related work examines repeated mailings (Bitran and Mondschein 1996), promotions during service interactions (Armony and Gurvich 2010), advertising intrusiveness (Goldfarb and Tucker 2011), creative and exposure histories (Braun and Moe 2013), and the match between retargeted content and customer interests (Lambrecht and Tucker 2013).

and states of Godfrey et al. (2011) and Todri et al. (2020). We randomize email frequency and directly observe opt-out, which differentiates our paper from the existing literature (see details in Table 1).

Research on permission-based marketing studies customers' decisions to enter or leave a communication relationship: Kumar et al. (2014) jointly model opt-in timing, opt-out timing, and spending, and Schweidel and Knox (2013) let direct marketing affect both transactions and the duration of an unobserved active relationship. Opt-out is an observed measure of disengagement, although it does not capture every form of annoyance; we report how the policy results change if subscribed customers also fatigue (see Sections 6.1 and 6.5).

2.2. Promotion Decisions and Customer Lifetime Value

Promotion research examines targeting and offers, from analytical and structural studies of targeted couponing under competition (Shaffer and Zhang 2002, Besanko et al. 2003) to personalized discounts and dynamic coupon policies (Jagabathula et al. 2022, Liu 2023), as well as how to schedule and value promotions (Baardman et al. 2019, Cohen et al. 2021, Baardman et al. 2023, Lin et al. 2024, Baek et al. 2025). Related empirical work documents behavioral biases in discretionary retail pricing (Liang et al. 2024) and strategic customer responses to discounted hotel upgrades (Yılmaz et al. 2022). Field studies show that promotion design affects behavior beyond the immediate purchase, through delayed incentives, category targeting, and discount depth (Kadiyala et al. 2024, Fong 2017, Reimers and Xie 2019, Anderson and Simester 2004). The catalog-mailing literature is the closest antecedent to our policy problem: Bitran and Mondschein (1996), Gönül and Shi (1998), and Simester et al. (2006) choose mailing frequency to maximize long-run value when mailings affect future purchases, and Simester et al. (2006) show in a field test how long-run effects complicate evaluating mailing policies. Catalog recipients cannot, however, opt out with a click, so the annoyance channel we study is absent there.

Beyond promotions, studies examine how website delays (Gallino et al. 2023) and offline return options (Nageswaran et al. 2025) shape customer behavior. Cui et al. (2024) optimize delivery promises to balance immediate sales against returns and retention.

The CLV literature helps weigh current revenue against future customer activity (Reinartz and Kumar 2003, Fader et al. 2005, Venkatesan and Kumar 2004, Gupta et al. 2006, Blattberg et al. 2009). Personalized communication and channel choice offer ways to influence that activity (Murthi and Sarkar 2003, Ansari and Mela 2003, Ansari et al. 2008). Our model draws on the representation of customer relationships as state transitions (Pfeifer and Carraway 2000). Li et al. (2026) optimize advertising campaigns under Markov customer dynamics and promotion fatigue, while Chang et al. (2026) combine a dynamic model with field experiments to balance immediate ad usage and future

retention in mobile games. Two strands inform our policy analysis. Ascarza (2018) shows that retention efforts should target customers by their sensitivity to the intervention rather than by their churn risk; our optimized policy has the same logic, targeting customers by their revenue and opt-out responses to frequency. Simester et al. (2020) show how targeting policies can be evaluated from one randomized experiment; in the same spirit, we evaluate segment rules on customers held out from their construction. Our contribution is to combine randomized evidence on frequency with historical estimates of behavior after opt-out. The experiment identifies short-run effects, while the historical estimates and their transfer to the experimental population depend on additional assumptions that we state and test.

3. Data and Empirical Setting

3.1. Setting and the Coupon-Email Program

Our partner is a major North American online retailer. The retailer sells furniture, home furnishings, and related household goods to consumers through its website and app.

The retailer's main promotional vehicle is a coupon email sent to subscribed customers, normally once a day. Each email contains a broadcast, storewide percentage discount (e.g., 15% off all products) without personalization, category restriction, or spending threshold; the discount depth varies by day between 12% and 25% and the coupon is valid for two days. Customers can leave the list through the unsubscribe link in the email footer or through the list-unsubscribe header; after opting out they receive no further coupon emails unless they resubscribe. Transactional emails, push notifications, on-site merchandising, and pricing are unaffected by subscription status. Opting out therefore removes the customer's access to both the reminder and the discount. Throughout, we use "opt-out" and "unsubscribe" interchangeably for leaving the coupon-email list (the retailer's interface says "unsubscribe").

We use two datasets. The first is a randomized field experiment, run during October 25–31, 2024, in which 603,153 subscribed customers were assigned to one of two email frequencies. For these customers, we also observe outcomes during the weeks before and after the experiment. The second is a historical panel that tracks 200,000 customers over the entire calendar year 2024. Both datasets record email frequency, unsubscriptions, and purchasing (revenue, item quantity, coupon usage) as well as auxiliary information like emails opened and static features such as join date. We define the main measures below:

- *Unsubscription.* In the experiment, unsubscription is a daily indicator equal to one if the customer opted out through a coupon email sent that day.
- *Email opens.* Opens are recorded by pixel-based tracking, which logs an image request when customers open an email.

- *Purchasing.* Revenue, units purchased, price per item, and coupon usage are measured from transaction records. Revenue is merchandise sales net of coupon and site-sale discounts. Price per item is revenue divided by units on customer-days with a purchase, and coupon usage is the dollar value of redeemed coupons.
- *Tenure.* Tenure always means time elapsed since a customer's first recorded visit; only the reference date changes across analyses. Tenure is recorded in days and converted to years where indicated.

The field experiment identifies the effects of email frequency on unsubscription and short-term revenue. The historical panel, introduced in Section 5, is used to estimate subsequent revenue and retention outcomes following unsubscription.

3.2. Field Experiment: Design and What Is Held Fixed

The experiment assigned 603,153 subscribed customers to daily coupon emails (control) or to an every-other-day schedule (treatment). Table 2 summarizes what varied and what remained fixed. Only the sending schedule differs: the offer, creative, send time, and unsubscribe mechanism are identical across arms. Because the coupon is valid for two days, a treated customer holds a valid coupon on most days of the week, so the arms differ in how many reminders arrive and only to a lesser extent in whether a discount is available.

Table 2 Field Experiment: Design Elements Held Fixed and Varied

Element	Treatment vs. control
Email frequency	**Varied**: daily (control) vs. every other day (treatment)
Coupon offer	Identical. Broadcast, storewide percentage discount (e.g., 15% off all products); not personalized, not category- or threshold-specific. Discount depth varied by day and took values of 12%, 15%, 17%, 20%, and 25%.
Coupon validity	Identical. Two days in both groups
Email creative	Identical. Same template, subject-line convention, imagery, and call-to-action
Send time	Identical. Same daily send window
Non-email channels	Identical. Transactional email, push notifications, on-site merchandising, and pricing were untouched
Unsubscribe mechanism	Identical. Same footer link and list-unsubscribe header in both arms

Table 3 compares arm-level purchase outcomes in the pre-experiment week; the differences are small and none is significant (smallest $p = 0.208$).

Table 3 Balance in the Week Before the Experiment (October 18–24, 2024)

	Revenue ($/Day)	Item Quantity	P(Purchase)	Price Per Item
Control	0.3555	0.0030	0.0017	132.09
Treated	0.3466	0.0029	0.0018	131.01
Rel. Size	-2.5%	-1.7%	+3.0%	-0.8%
t-test p-value	0.597	0.703	0.208	0.809

Daily averages per customer over the seven pre-experiment days. Rel. Size is the treated-minus-control difference as a share of the control mean.

3.3. Model-Free Observations from the Field Experiment

We consider our data of the 603,153 customers from the week before, week of, and week after the randomized experiment. Figure 1 plots daily opt-out rates, revenue, and purchase rates by treatment arm before, during, and after the experimental week, with 95% confidence bands.

The outcomes track closely before the experiment, separate while the two email schedules are in place, and move back together after the experiment ends. Relative to the daily-email control, the every-other-day schedule reduces the opt-out rate by 59% and average daily revenue by 8% during the experimental week. The red dashed line marks October 25, the first day of treatment, so the gap on that date is part of the treatment-period difference rather than evidence of a pre-existing imbalance.

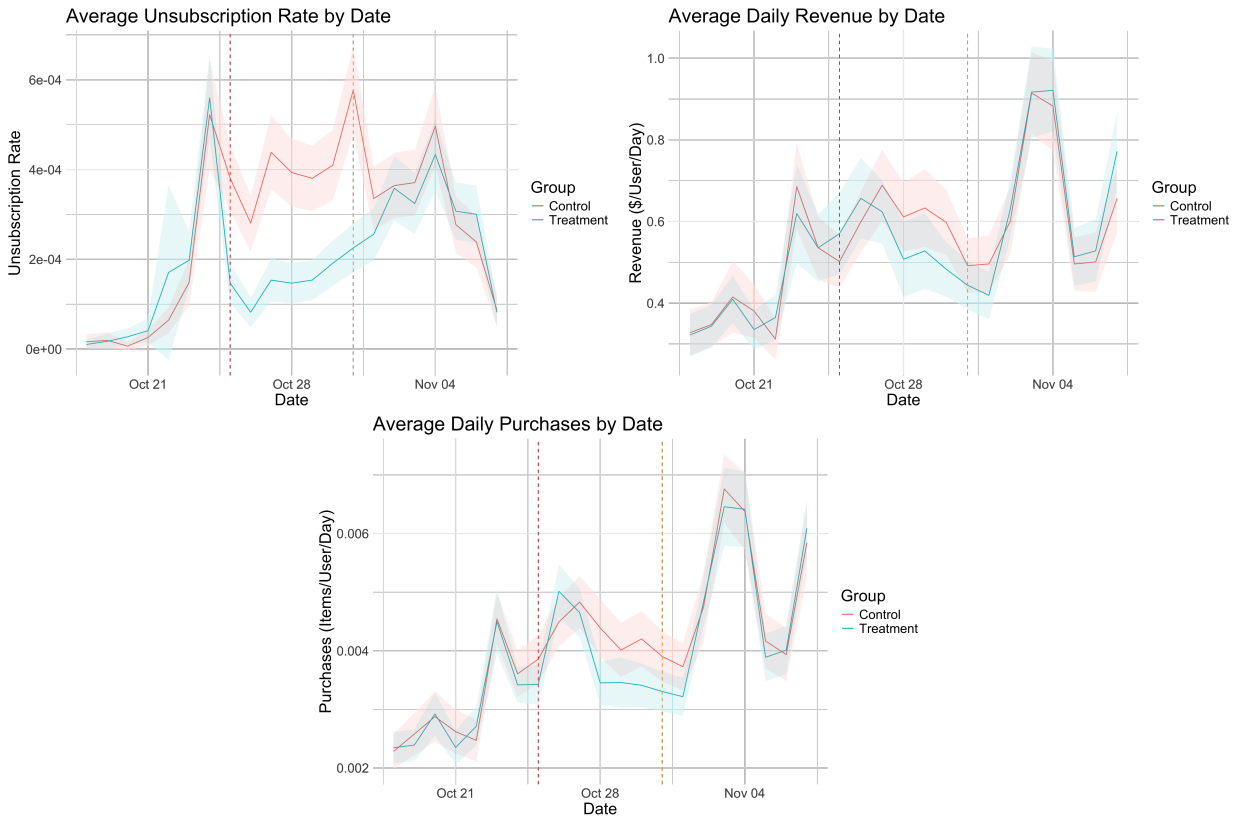


Figure 1 Daily opt-out rate, revenue, and purchase rate by arm before, during, and after the experiment, with 95% confidence bands. The red dashed line marks the first day of the experiment (October 25, 2024) and the orange line its last day (October 31); the post-period ends on November 7.

As seen from Figure 1, email frequency directly affects both purchasing and opt-out. Each email reminds customers of the retailer and makes the coupon offer salient, increasing short-run purchases; in this sense, discount offers act partly as advertising (Sahni et al. 2017). Repeated messages, however, also impose attention costs and can generate wearout and reactance (Pechmann

and Stewart 1988, Brehm 1966, Edwards et al. 2002), increasing the likelihood of opt-out. Once customers opt out, the loss of future reminders and coupons may reduce subsequent purchasing and, ultimately, retention.

4. Field Experiment: Effects of Email Frequency

4.1. Main Effects

Our baseline specification uses randomization to estimate how reduced frequency affects the outcome of customer i on day t:

$$\text{Outcome}_{it} = \beta\,\text{Treated}_i + \delta_t + \varepsilon_{it}, \tag{1}$$

where δ_t denotes date fixed effects and ε_{it} is the error term. Random assignment supports interpreting β as the average effect of assignment to every-other-day emails. The outcomes are the daily opt-out indicator, revenue, item quantity, purchase incidence, and price per item (observed only on purchasing customer-days).

Table 4 reports the estimates, the control-group means, and 95% confidence intervals for absolute and relative effects. Absolute coefficients are small because they are measured per customer-day at low base rates; the control group opts out at a rate of 0.034% per day (0.24% per week) and purchases on 0.23% of days. The estimates establish the trade-off:

- *Opt-out.* Every-other-day emails reduce the daily opt-out probability by 0.020 percentage points, corresponding to a 59% reduction relative to the control mean, with a 95% confidence interval of 51–68%. If the daily opt-out rates were to persist for a full year, under the constant-rate projection in Section 6.1, the annual opt-out rate would decline from 11.8% to 5.0%.
- *Purchases and revenue.* Fewer emails reduce purchase incidence by 4.8% (95% confidence interval: 1–9%), item quantity by 10% (95% confidence interval: 5–15%), and revenue by 7.6% (95% confidence interval: 0.2–14.9%). The revenue effect is statistically significant at the 5% level ($p = 0.043$). In economic terms, it corresponds to approximately \$0.26 per customer per week, or about \$75,000 in weekly revenue across the treatment group. We find no statistically significant effect on price per item.

Appendix A reports logistic and Poisson models and specifications with log-transformed outcomes. The opt-out, item-quantity, and purchase-incidence effects remain negative and statistically significant; the revenue effect is also negative and significant with log-transformed revenue.

Because treated customers hold a valid two-day coupon on most days, the treatment primarily changes reminder frequency, although it also creates some differences in discount availability. The resulting revenue decline therefore largely reflects the value of promotional reminders for customers who purchase infrequently, consistent with evidence that discount emails operate partly as advertising (Sahni et al. 2017). At the same time, part of the effect may reflect days on which treated

Table 4 Average Treatment Effects of Reducing Email Frequency from Daily to Every Other Day

	Dependent variable:				
	P(Unsubscribe)	Revenue ($/Day)	Item Quantity	P(Purchase)	Price Per Item
	(1)	(2)	(3)	(4)	(5)
Treated	−0.000204***	−0.037**	−0.00042***	−0.00011**	0.671
	(0.000015)	(0.018)	(0.00011)	(0.00005)	(4.021)
Control mean	0.00034	0.48	0.0042	0.0023	129
95% CI (absolute)	[−0.00023, −0.00017]	[−0.072, −0.001]	[−0.00065, −0.00020]	[−0.00020, −0.00002]	[−7.2, 8.6]
Rel. Effect Size	−59.2%	−7.6%	−10.0%	−4.8%	+0.5%
95% CI (relative)	[−67.9%, −50.5%]	[−14.9%, −0.2%]	[−15.3%, −4.7%]	[−8.7%, −1.0%]	[−5.6%, 6.6%]
Observations	4,222,071	4,222,071	4,222,071	4,222,071	9,635
Fixed Effects	Date	Date	Date	Date	Date

Note: *p<0.1; **p<0.05; ***p<0.01

Estimates use the experimental week. Price per item is observed only on purchasing customer-days. Control means are daily rates in the control group during the experimental week; relative effects divide the treatment coefficient and its confidence limits by the control mean.

customers do not hold a valid coupon. Fewer emails also mechanically create fewer opportunities to click "unsubscribe." We view this as part of the treatment rather than a confound, because the relevant object for the CLV model is the total opt-out probability under each email schedule, regardless of the immediate mechanism generating it. Section 4.2 examines whether the opt-out response varies with recent purchase activity and discusses what this implies for customers who may be completing a purchase cycle.

Both groups resumed daily emails on November 1. Comparisons by original treatment assignment for November 1–7, estimated with date fixed effects, show no differences that are statistically significant at the 5% level: daily revenue is +$0.018 per customer-day (standard error 0.019; 95% confidence interval −$0.019 to $0.054; control mean $0.528), item quantity is −0.00009 (0.00014), purchase probability is +0.00009 (0.00005, $p = 0.06$), and opt-out is +0.000007 (0.000016). The revenue confidence interval includes zero and positive effects as large as $0.054 per customer-day, so these estimates do not establish the absence of rebound or carry-over effects. They also cannot address fatigue that may accumulate over a longer period of exposure. Section 6.1 discusses the implications of extrapolating from a one-week experiment to a sustained email-frequency policy.

4.2. Heterogeneous Effects by Customer Characteristics

The reduction in opt-out and the short-run revenue cost of reduced email frequency may be concentrated among different customers. We examine three customer characteristics. *Tenure* captures relationship length: promotions affect new and established customers differently (Anderson and Simester 2004), and customers whose relationship with the retailer is less established should show a larger opt-out response. *Recent purchase activity* (item quantity in the past seven days) captures whether the customer values the offer: active customers should respond to reminders but have little reason to opt out under either schedule. *Email engagement* (recorded opens in the past seven days) captures whether the customer sees the messages: an email's reminder value and attention

cost accrue only to customers who open it, so openers should show a larger opt-out response to frequency and a smaller revenue response, since they continue to see offers with every-other-day emails. All three moderators are measured before treatment, with tenure evaluated on October 25, 2024, and recent purchase activity and email engagement measured over October 18–24. For each moderator M_i, we estimate

$$\text{Outcome}_{it} = \beta\,\text{Treated}_i + \gamma\,M_i + \theta\left(\text{Treated}_i \times M_i\right) + \delta_t + \varepsilon_{it}, \tag{2}$$

and a combined model that includes all three moderators and their interactions. The coefficient θ is the change in the treatment effect per unit of M_i. Random assignment supports treatment comparisons within pre-treatment groups; the linear interaction imposes a functional form. Table 5 reports the combined model, which is our preferred specification because the moderators are correlated. Appendix C reports the single-moderator models and Appendix B reports median-split versions; the interactions significant at the 5% level have the same signs in all versions, although the tenure interaction in the opt-out equation is weaker with a median split.

In Table 5 and the corresponding experimental heterogeneity tables in Appendices B and C, opt-out coefficients and their standard errors are multiplied by 100 and reported in percentage-point units.

Table 5 Heterogeneous Treatment Effects: Combined Specification

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Tenure (years)	−0.00087***	0.0068**
	(0.00023)	(0.0028)
Item Quantity	0.00330	6.632***
	(0.00249)	(0.029)
Email Opens	0.01123***	0.007
	(0.00083)	(0.010)
Treated	−0.01756***	0.042
	(0.00321)	(0.038)
Tenure×Treated	0.00074**	0.0047
	(0.00034)	(0.0040)
Item Quantity×Treated	0.00175	−6.189***
	(0.00485)	(0.057)
Email Opens×Treated	−0.00939***	0.015
	(0.00119)	(0.014)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date

Note: *p<0.1; **p<0.05; ***p<0.01
Item quantity and email opens are counts over the seven days before the experiment.

Customer tenure. The tenure×treated interaction in the opt-out equation is positive and significant: the opt-out reduction from fewer emails diminishes with tenure, so newer customers show the largest reduction. The interaction in the revenue equation is small and insignificant. The tenure interaction in the opt-out equation is the least robust: with a median split it is significant only at the 10% level in the combined model and not on its own (Appendix B).

Recent purchase activity. The item-quantity interaction dominates the revenue equation: each additional item bought in the pre-experiment week raises the revenue loss from reduced frequency by \$6.19 per day, and the median-split version shows an additional loss of \$1.25 per day for customers with any recent purchase, compared with no significant loss for the rest (Table 19). The opt-out interaction is insignificant. Recent buyers thus respond to reminders, but their opt-out treatment effect does not differ significantly from that of other customers. This concentration of the revenue response helps make personalization valuable in Section 6, where the model combines revenue and opt-out responses to assign frequency.

Email engagement. The opens×treated interaction in the opt-out equation is negative and significant: the opt-out reduction from fewer emails is larger for customers who open more emails, consistent with an email's attention cost accruing only to customers who read it. On the revenue side, the interaction is positive but insignificant in the combined model (significant at the 10% level in the single-moderator model), consistent with openers still seeing the offer with every-other-day emails. Customers who rarely open emails show a smaller opt-out reduction; the revenue interaction is imprecise.

These results provide no clear evidence that purchase-cycle completion alone explains the experimental opt-out effect, as the interaction between treatment and recent purchasing is not statistically significant for opt-out. By contrast, the revenue response is concentrated among recent purchasers, although these patterns do not by themselves identify the underlying mechanism. The heterogeneity results in Table 5 therefore describe which customers respond differently to email frequency. Whether the consequences of opting out also vary across customers is a distinct question, which we examine in Section 5.4.

5. Long-Run Consequences of Opt-Out

To assess the implications of unsubscription for customer lifetime value, we use the historical panel to estimate how opting out is associated with subsequent spending and retention. The panel covers 200,000 customers randomly sampled from those with at least one purchase in 2024; the retailer drew the historical and experimental samples independently. The central identification challenge is endogeneity: opt-out is a customer choice, and the same underlying processes that lead a customer to unsubscribe, such as completing a purchase cycle or losing interest in the retailer, may also reduce subsequent spending. We therefore report several estimators that address this concern in different ways and carry the resulting range of estimates into the policy analysis.

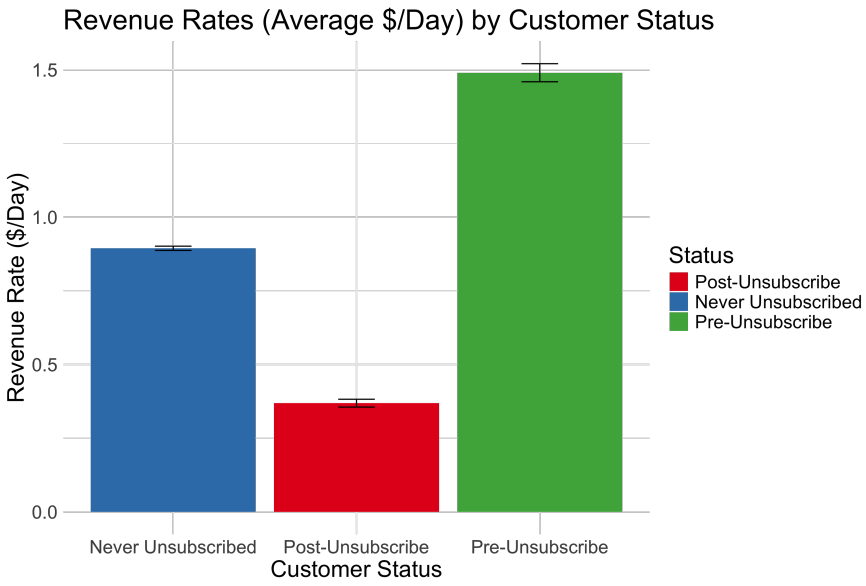


Figure 2 Daily revenue before and after opt-out in the historical panel. Spending is elevated before opt-out and falls sharply afterwards; Section 5.2 examines this association using regression and instrumental-variable specifications.

5.1. Model-Free Evidence of Revenue Difference Around Opt-Out

We first examine the raw spending patterns around opt-out to motivate the revenue regressions that follow. Figure 2 plots average daily revenue around the opt-out date in the historical panel. After opting out, customers spend about three-fifths less per day than customers who never opt out and three-quarters less than they themselves spent before opting out. Before opting out, however, their spending is *above* that of customers who never opt out. This pre-opt-out spike matters: a share of opt-outs follow a purchase, consistent with customers who buy what they need and then leave the list.

Opt-out may reflect a decline in purchase intent that is already underway. Customers may subscribe to meet a temporary need and unsubscribe after purchasing, or they may gradually lose interest, stop engaging with emails, and eventually opt out. In either case, spending may fall after opt-out partly for reasons that predate it. The pattern in Figure 2 could therefore arise even if losing access to emails had no causal effect on spending. The raw before–after gap may thus may overstate the decline attributable to losing access to emails. Section 5.2 presents regression and instrumental-variable estimates that address this concern in different ways; these estimates are used in the CLV policy model.

5.2. Revenue After Opt-Out

A customer-month enters the revenue regressions if the customer was still subscribed at the start of the month, received at least one coupon email during the month, and is observed next month.

The outcome is next-month revenue, so December is excluded as a predictor month. These rules yield 1,095,832 customer-month observations. The instrumental-variable (IV) specification additionally requires the preceding month's email opens, giving 1,016,252 observations with predictor months February–November. The retention regressions use separate customer-level cross-sections containing 23,465 and 118,941 observations, described in Section 5.3.

Specification. Using customer-month observations, we estimate

$$\text{Revenue}_{it} = \beta\,\text{Unsubscribed}_{i,t-1} + \phi\,\text{Revenue}_{i,t-1} + \mu_t + \varepsilon_{it}, \tag{3}$$

where Revenue_{it} is customer i's net revenue in month t, $\text{Unsubscribed}_{i,t-1}$ indicates the first recorded opt-out in month $t-1$, $\text{Revenue}_{i,t-1}$ is the previous month's revenue, and μ_t are calendar-month fixed effects that absorb seasonality in spending. The estimation sample consists of the customer-months described above; observations whose predictor month $t-1$ falls after the opt-out month are excluded, so β compares the month immediately after opt-out with months in which the customer was subscribed. Lagged revenue controls for the customer's recent spending level. Because purchase intent may change alongside opt-out, we complement this specification with IV and within-customer estimates.

Instrumental variable. We instrument $\text{Unsubscribed}_{i,t-1}$ with the change in recorded customer email opens from month $t-2$ to month $t-1$, $z_{i,t-1} = \text{Opens}_{i,t-1} - \text{Opens}_{i,t-2}$:

$$\text{Revenue}_{it} = \beta\,\widehat{\text{Unsubscribed}}_{i,t-1} + \alpha_i + \mu_t + \varepsilon_{it}, \tag{4}$$

where $\widehat{\text{Unsubscribed}}_{i,t-1}$ is the first-stage fitted value from regressing the opt-out indicator on the change in opens and on customer and month fixed effects. The IV sample contains 1,016,252 customer-months from 153,476 customers, including 27,927 opt-outs. The instrument is strongly relevant: one additional recorded email open relative to the preceding month is associated with a 0.115-percentage-point reduction in the probability of opting out (Table 29; partial $F = 20{,}225$). Under the exclusion restriction and the standard IV assumptions, the resulting IV estimate identifies the effect of opt-out for customers whose decision to unsubscribe responds to changes in email engagement (Angrist et al. 1996).

The exclusion restriction requires that, conditional on customer and month fixed effects, changes in recorded email opens in the previous month do not directly affect revenue in the current month.[3] Appendix D reports the first-stage results and additional robustness checks, while Appendix E presents complementary within-customer estimates.

[3] Coupons distributed by email are valid for only two days, which limits the scope for direct carry-over of discounts across months. However, short coupon validity does not rule out other channels, such as changes in purchase timing or broader shifts in customer engagement. We therefore view the IV estimates as suggestive evidence rather than as fully causal estimates.

Results. Table 6 reports the OLS and IV estimates. Opt-out is associated with a \$9.85 decline in next-month revenue in the OLS specification and a \$9.31 decline in the IV specification, corresponding to 44% and 41% of the panel-wide monthly mean of \$22.47, respectively. Appendix E reports complementary within-customer estimates in Table 30: the estimated decline is \$5.83 with customer and month fixed effects and \$7.93 when lagged revenue is added. We do not include lagged revenue in the main-text IV specification because, in a short panel with customer fixed effects, a lagged dependent variable can induce bias (Nickell 1981). Appendix E illustrates this issue through the sign reversal on the lagged-revenue coefficient. We view the within-customer estimates as complementary evidence on the post-opt-out decline. Across the OLS, IV, and within-customer specifications, the estimated reduction in monthly spending ranges from roughly \$5 to \$10. These estimates feed into the CLV model in Section 6, where the sensitivity analysis scales the historical inputs by factors between 0.5 and 1.5 to span this range.

Table 6 Revenue in the Month After Opt-Out

	Dependent variable:	
	Month Revenue (\$)	
	(1)	(2)
Previous Month Revenue	0.086***	
	(0.001)	
Previous Month Unsubscribed	−9.853***	−9.313***
	(0.745)	(2.978)
Observations	1,095,832	1,016,252
Opt-outs	29,103	27,927
Fixed Effects	Month	Customer, month
Estimator	OLS	IV

Note: *p<0.1; **p<0.05; ***p<0.01
Net revenue in month t regressed on month $t-1$ variables; the sample excludes observations whose predictor month $t-1$ falls after the opt-out month. Column (2) instruments opt-out with the change in email opens (first stage in Table 29).

5.3. Retention After Opt-Out

We estimate

$$\text{Survival}_i = \alpha + \beta\,\text{Unsubscribed}_i + \gamma\,\text{Tenure}_i + \theta\,(\text{Unsubscribed}_i \times \text{Tenure}_i) + \varepsilon_i \qquad (5)$$

using two customer-level cross-sections, in both of which Unsubscribed_i indicates permanent opt-out:

- **Column (1), "After Month 1".** The sample consists of the 23,465 customers who purchased in month 1. Opt-out is measured in month 1, and survival indicates at least one purchase during months 2–12.

- **Column (2), "After Months 1–6".** The sample includes the 118,941 customers who purchased during months 1–6; it is larger than the first because it pools six months of purchasers rather than one. Opt-out is measured during months 1–6, and survival indicates a purchase during months 7–12.

Each customer contributes a single observation, so customer fixed effects are not available. In column (1), where every opt-out occurs in month 1, a month fixed effect would also be perfectly collinear with the constant. We therefore control for observed relationship length using tenure and its interaction with opt-out. Table 7 reports the main estimates, while Appendix A presents the corresponding logistic specifications. Because tenure enters the regression without centering, the opt-out coefficients correspond to zero tenure. Evaluated at each sample's observed tenure distribution, the average predicted opt-out contrasts are -0.156 and -0.066. In raw terms, 7.7% of the 326 permanent opt-outs in column (1) made another purchase during months 2–12, compared with 23.9% of other customers. In column (2), 4.3% of the 12,466 opt-outs purchased during months 7–12, compared with 11.1% of other customers. The cohort analysis described below uses a narrower definition, counting only opt-outs occurring in the purchase month, and therefore includes fewer opt-outs. Overall, opting out is followed by a decline of more than half in the probability of remaining an active buyer.

A customer who opts out in January has a longer follow-up period than one who opts out in June, and the reasons for opting out may also vary over the course of the year. To examine whether retention depends on the timing of opt-out, Appendix E (Table 31) repeats the analysis separately for each opt-out month $m = 1, \ldots, 6$ using a common six-month horizon. For each month m, the risk set consists of customers who purchased in month m and were subscribed at the beginning of that month, the opt-out indicator identifies customers who permanently opted out during month m, and survival is defined as making a purchase during months $m+1$ through $m+6$. Across all cohorts, six-month survival ranges from 6% to 10% among customers who opt out and from 17% to 19% among other customers. The cohort-specific coefficients range from -0.061 to -0.094, while the pooled specification with cohort fixed effects yields an estimate of -0.076 (standard error 0.008). We also fail to reject equality of the coefficients across cohorts ($p = 0.28$). The CLV model uses annual retention probabilities that do not vary with the month of opt-out, and the stability of these estimates across cohorts provides support for this simplifying assumption.

5.4. Heterogeneity in the Consequences of Opt-Out

Section 4.2 showed that the effect of email frequency on *opt-out propensity* varies with tenure and email engagement. A separate question is whether the *consequences* of opting out also vary across customers. Table 26 in Appendix C interacts the opt-out indicator in the OLS specification with the

Table 7 Retention After Opt-Out

	Dependent variable:	
	Survival	
	After Month 1	After Months 1–6
	(1)	(2)
Unsubscribed	−0.116***	−0.051***
	(0.030)	(0.004)
Tenure (years)	0.0109***	0.0056***
	(0.0005)	(0.0002)
Unsubscribed×Tenure	−0.0077	−0.0031***
	(0.0058)	(0.0006)
Constant	0.183***	0.084***
	(0.004)	(0.001)
Observations (customers)	23,465	118,941
Opt-out window	Month 1	Months 1–6
Survival window	Months 2–12	Months 7–12
Mean survival, opt-outs / others	0.077 / 0.239	0.043 / 0.111
Average predicted contrast	−0.156	−0.066

Note: *p<0.1; **p<0.05; ***p<0.01
Each customer contributes one observation. Survival indicates at least one purchase in the stated window. Tenure is uncentered, so the opt-out coefficient applies at zero tenure; the average predicted contrast evaluates the model at each sample's tenure distribution.

same three moderators. The interaction with item quantity is negative and statistically significant, whereas the interactions with tenure and email opens are not. When the moderators are instead defined using median splits, none of the interactions is statistically significant (Table 22). Because these results are sensitive to the moderator specification, the policy model assumes a common post-opt-out revenue multiplier across customers, while allowing retention probabilities to vary with tenure.

6. Optimizing Email Frequency for Customer Lifetime Value

The experiment shows how email frequency affects short-run purchasing behavior and opt-out, while the historical panel provides evidence on the longer-run consequences of opting out. This section brings these two pieces together in a model of the retailer's decision problem and uses it to compare alternative email policies. Section 6.1 defines the decision problem and model inputs, Section 6.2 describes the policies we evaluate and how they are computed, and Sections 6.3–6.6 present the results, examine which customers should continue receiving daily emails, assess the sensitivity of the conclusions, and discuss the resulting managerial implications.

6.1. Decision Problem and Model Inputs

The retailer assigns each customer i, characterized by X_i, a permanent email frequency $y_i \in \{0, 1\}$, where $y_i = 1$ denotes daily emails (the status quo) and $y_i = 0$ every-other-day emails, to maximize

the customer's expected lifetime revenue. We take a year as the planning period and describe each customer by six quantities, summarized in Table 8:

- $u_i(y)$, the probability that the customer opts out during a year of frequency y;
- $\rho_i(y)$, the customer's expected annual revenue while subscribed under frequency y;
- m, the fraction of that revenue the customer continues to generate after opting out, so that revenue after opt-out is $m\,\rho_i(y)$;
- s_i^N and s_i^U, the probabilities that the customer purchases again next year (retention) if the customer remains subscribed (N) or opts out (U);
- $c_i(y)$, the customer's expected annual coupon redemptions under frequency y, used only in the budget constraint.

Frequency-dependent quantities come from the experiment. We predict $u_i(y)$, $\rho_i(y)$, and $c_i(y)$ for both frequencies using linear models with frequency interactions, as in Section 4.2. Revenue predictions use tenure, revenue over the preceding 30 days, recorded email opens over the preceding 14 days, and item quantity and coupon usage over the preceding seven days; opt-out and coupon-usage predictions use the first three covariates. We convert daily rates to annual quantities: daily revenue and coupon usage are multiplied by the number of days in the year, and the daily opt-out probability is compounded over the year, which treats opt-out as a memoryless daily event. The policy considers only the two email frequencies directly tested in the experiment. Extending their estimated effects over longer horizons and transferring the historical estimates to the experimental population therefore require the assumptions stated below. The quantities governing post-opt-out outcomes are estimated from the historical panel: the post-opt-out revenue multiplier m is based on the decline in revenue following opt-out (Section 5.2), while s_i^N and s_i^U come from the retention regressions in Section 5.3, where retention depends on customer tenure and opt-out status.

Table 8 Model Inputs and Their Sources

Quantity	Meaning	Depends on	Estimated from
$u_i(y)$	Probability of opting out within a year	X_i, y	Experiment, interaction model of Section 4.2
$\rho_i(y)$	Annual revenue while subscribed	X_i, y	Experiment, interaction model of Section 4.2
$c_i(y)$	Annual coupon redemptions	X_i, y	Experiment, interaction model of Section 4.2
m	Revenue after opt-out relative to before	–	Historical panel, Section 5.2
s_i^N, s_i^U	Retention if subscribed / if opted out	Tenure	Historical panel, Section 5.3

Given these inputs, a customer active at year-start under frequency y generates expected revenue

$$R_i(y) \;=\; \big[1-u_i(y)\big]\,\rho_i(y) \;+\; u_i(y)\,m\,\rho_i(y) \tag{6}$$

during the year and remains active into the following year with probability

$$S_i(y) \;=\; \big[1-u_i(y)\big]\,s_i^N \;+\; u_i(y)\,s_i^U. \tag{7}$$

Treating every year in which the customer remains active as a repetition of the first, and discounting at an annual rate r, the customer's lifetime value under frequency y is

$$CLV_i(y) = \sum_{k\geq 1} \frac{R_i(y)\, S_i(y)^{k-1}}{(1+r)^k} = \frac{R_i(y)}{1+r-S_i(y)}. \tag{8}$$

This is the constant-retention form of customer lifetime value (Gupta et al. 2006), or equivalently a two-state Markov chain with an active and an inactive state (Pfeifer and Carraway 2000). With $r=0$ it is annual revenue while active, $R_i(y)$, times the expected number of active years, $1/(1-S_i(y))$. Frequency enters through two channels that pull in opposite directions: daily emails raise ρ_i, because reminders trigger purchases, and raise u_i, which lowers both R_i and S_i. Revenue is net of coupon and site-sale discounts, so CLV_i measures net revenue before the cost of goods.

Assumptions. Five assumptions connect the experimental and historical estimates to the lifetime-value calculation in (6)–(8). The first links the experiment to the long-run model.

ASSUMPTION 1 **(Opt-out sufficiency)**. *Conditional on X_i and on whether the customer opts out, retention does not depend on email frequency: s_i^N and s_i^U are the same under $y=0$ and $y=1$.*

Under this assumption, frequency affects retention only through opt-out. A customer who stays subscribed under daily emails may nonetheless tire of them beyond the one-week experiment. Section 6.5 therefore recomputes the policies with the retention of subscribed daily-email customers reduced by $\delta \in \{0.01, 0.02, 0.05\}$; such fatigue raises the value of reduced frequency, so the baseline results are conservative in this respect.

The second assumption is stationarity: the revenue and retention functions in (6)–(7) apply to each year in which the customer remains active. Subscription status is not carried into later years, so customers who opt out but remain active are assigned the same rates as customers who begin the year subscribed.

The third is that the post-opt-out multiplier m is common to all customers, while retention varies with tenure; Section 5.4 explains why we do not let m vary.

The fourth is transfer: m and the retention probabilities are estimated in the historical panel and are applied, conditional on tenure, to the experimental population.

The fifth is that behavioral covariates are fixed at their pre-experiment values, with tenure measured on October 31, 2024, and the frequency is chosen once. Recent purchase activity is transient, so the rule "daily emails for recently active customers" that emerges in Section 6.4 could be implemented as a rolling rule applied whenever a customer has just purchased; the static model values it as if the recently active customers of October 2024 remained so, which may overstate the loss from moving them to every other day.

From a one-week experiment to a sustained policy. The experiment measures responses to a one-week change in email frequency. Applying these responses to sustained policies assumes that they remain informative over a longer horizon; two observations help assess the reliability of the one-week estimates. First, the post-experiment week shows no statistically significant rebound or carry-over in revenue or opt-out (Section 4.1). Second, the policy gains move smoothly with the size of the inputs (Section 6.5), so whether responses attenuate or accumulate determines the direction of the error. A longer follow-up would allow the persistence of these responses to be assessed directly.

Discount rate. We report $r = 0$ as the baseline because retention already discounts the future heavily: expected revenue next year is weighted by the retention probability, 0.376 on average, so the percentage gains of the personalized policies are relatively insensitive to financial discounting, even though a 10% annual discount rate lowers status-quo CLV by 14.3%. Section 6.5 reoptimizes at rates up to 10%: the percentage gains are nearly unchanged and at most 0.01% of customers change assignment.

6.2. Policies and Solution Approach

We compare four policies. The *status quo* sends daily emails to everyone. The *uniform every-other-day* policy halves frequency for everyone. The *myopic* policy chooses y_i to maximize the customer's revenue rate while subscribed, $\rho_i(y_i)$, and ignores opt-out; it evaluates each frequency on campaign revenue alone, as a one-week test would, and may assign a lower frequency when it predicts higher short-run revenue. The *CLV-optimal* policy chooses y_i to maximize $CLV_i(y_i)$. For the two personalized policies, we solve

$$\max_{y\in\{0,1\}^N} \sum_{i=1}^{N} V_i(y_i) \qquad \text{s.t.} \qquad \sum_{i=1}^{N} c_i(y_i) \le B, \tag{9}$$

where V_i is CLV_i or ρ_i; we solve each problem both without the budget constraint and with B equal to projected annual coupon usage under the status quo. The retailer manages promotional discount dollars as a budget line, and, as Section 6.3 shows, the personalized policies raise predicted coupon redemptions, so the constrained version answers the question a budget-constrained retailer would ask.

Without the budget, the problem separates across customers: each receives the frequency with the higher predicted objective, one comparison per customer. With the budget, it is a 0–1 knapsack problem, which we solve by Lagrangian relaxation. Let Δ_i^v and Δ_i^c denote the changes in predicted objective and coupon usage when customer i is switched from daily to every-other-day emails. Customer i switches when $\Delta_i^v - \lambda\,\Delta_i^c > 0$ for budget multiplier $\lambda \ge 0$, and λ is raised until total

coupon usage no longer exceeds B. The gap between the resulting objective and the relaxation bound equals λ times the unused budget, which is numerically negligible in our application.[4]

6.3. Results: Unconstrained and Constrained Policies

Table 9 Predicted Outcomes Under Uniform and Personalized Email Policies

Email Policy	CLV ($)	Revenue Rate ($/Month)	E[Revenue] ($/Year)	Retention (/Year)	Coupon Usage ($/Year)	Schedule (Emails/Day)
Daily (status quo)	213.03	11.78	130.3	0.376	12.42	1
Every other day	213.56	10.94	127.2	0.388	11.55	0.50
Myopic	227.46	12.12	136.9	0.382	17.61	0.68
CLV-optimal	230.78	12.04	138.2	0.388	19.85	0.52

Averages over the 603,153 experimental customers. Revenue Rate is monthly revenue while subscribed, $\rho_i(y_i)/12$; E[Revenue] is $R_i(y_i)$ in Equation (6); Retention is $S_i(y_i)$ in Equation (7); CLV is Equation (8) with $r = 0$; Schedule is the average number of emails per day (1 for daily, 0.5 for every other day).

Table 9 reports the results without a coupon budget. Under the status quo, average CLV is \$213.03. A uniform switch to every-other-day emails leaves average CLV nearly unchanged (\$213.56, +0.25%): the retention gain offsets the revenue loss on average, while coupon usage falls by 7%.

Personalized policies yield larger predicted gains. The myopic policy raises CLV to \$227.46 (+6.8%), through higher annual revenue (+5.1%). The CLV-optimal policy raises CLV to \$230.78 (+8.3%, or 1.5% more than the myopic policy) by accepting a slightly lower revenue rate while subscribed to gain higher retention, and it assigns every-other-day emails to more customers than the myopic policy (average schedule 0.52 vs. 0.68 emails per day).

Both personalized policies raise predicted coupon usage, from \$12.42 per customer-year under the status quo to \$17.61 under the myopic and \$19.85 under the CLV-optimal policy, even though they send fewer emails. The prediction model attributes to customers without recent purchases a small positive response of coupon redemption to every-other-day emails, while most recent purchasers, who redeem most coupons, are kept on daily emails. Table 11 shows that predicted coupon usage rises in all eight descriptive cells. The constrained analysis in Table 10 prevents an increase in predicted aggregate coupon redemptions, although its assignments still depend on the redemption model.

With coupon usage capped at \$12.42 per customer-year, the myopic policy gains 2.3% and the CLV-optimal policy 4.5%. The CLV-optimal policy gains more under the same budget because it

[4] On the 603,153-customer file, computing the predictions and searching for the multiplier takes under one second. For the CLV objective the budget binds to within \$0.00001 per customer-year, and the relative optimality gap is 7×10^{-8}. With additional email frequencies, the problem becomes a multiple-choice knapsack (Sinha and Zoltners 1979, Kellerer et al. 2004) and the same multiplier search applies; what limits the menu is estimation, since each option needs its own experimental arm (as in the frequency design of Sahni et al. 2019). Because the annoyance response may be nonlinear (Godfrey et al. 2011), we caution against interpolating between tested frequencies.

Table 10 Predicted Outcomes Under Personalized Policies with Coupon Usage Held at the Status-Quo Level

Email Policy	CLV ($)	Revenue Rate ($/Month)	E[Revenue] ($/Year)	Retention (/Year)	Coupon Usage ($/Year)	Schedule (Emails/Day)
Myopic	218.02	11.87	132.1	0.377	12.41	0.88
CLV-optimal	222.54	11.65	133.2	0.385	12.42	0.63

Columns as in Table 9. Expected coupon usage may not exceed its status-quo level of $12.42 per customer-year.

allocates redemption budget where an avoided opt-out is worth most. The constrained policies differ from the unconstrained ones: the CLV-optimal policy now keeps daily emails for about a quarter of customers rather than 5% (an average schedule of 0.63 vs. 0.52 emails per day), because moving customers without recent purchases to every other day is predicted to raise their redemptions. Section 6.5 shows that both sets of gains are robust to discount rates, opt-out-sufficiency departures, and historical inputs.

6.4. Who Should Receive Daily Emails?

The CLV-optimal policy uses each customer's covariates, but its structure is simple. Table 11 crosses median splits of tenure, past-year spending, and past-year coupon usage and reports, for the eight cells, the average CLV under each uniform policy and the average email schedule under the individual (CLV-optimal) policy.

Within each cell, most customers are assigned every-other-day emails: about 95–97% have higher predicted CLV under this schedule. Where cell averages favor daily emails, they reflect the 5% of recent buyers, for whom the predicted loss from fewer emails is large. Among the 94.9% of customers with no revenue in the past 30 days and no items in the past 7 days, every-other-day emails yield the higher predicted CLV for all but three customers; among the 30,485 recently active customers, daily emails do for 96.3%. This is the experimental heterogeneity of Section 4.2 carried through the model: the model predicts that the revenue benefit of daily reminders outweighs their opt-out cost for recent buyers, whereas for other customers the modeled retention benefit favors fewer emails. The large individual differences for recent buyers arise because the week-after-purchase revenue response is annualized; they identify who should keep daily emails rather than the dollar value of doing so.

On their own, the three splits in Table 11 provide limited guidance for choosing email frequency. When each cell's preferred uniform frequency is estimated on half the customers and evaluated on the remainder, the cell rule captures only about one-sixth of the gain of the individual policy (1.4% vs. 8.3% over the status quo). By contrast, a two-segment rule that sends daily emails to recently active customers and every-other-day emails to everyone else attains $230.76, essentially the individual policy's $230.78. Recent activity, rather than tenure, annual spending, or coupon

usage, separates the customers for whom daily emails are worth their annoyance cost. These assignments are for the unconstrained problem; with coupon redemptions capped, the CLV-optimal policy keeps daily emails for about a quarter of customers (Table 10), and we have not characterized that assignment by segment. Appendix F reports the uniform-policy comparisons for each split.

Table 11 Predicted CLV and Assignments by Tenure, Past-Year Spending, and Past-Year Coupon Usage

Tenure	Spending	Coupons	Customers	CLV, uniform policy ($)		Individual policy	Coupon usage ($/year)		Cell rule
>2,583 d	>$179	>$17		Daily	Every other day	schedule (emails/day)	Daily	Individual	
No	No	No	151,627	183.73	179.22	0.526	6.67	13.25	Daily
No	No	Yes	7,433	168.16	173.82	0.517	4.26	11.05	Every other day
No	Yes	No	88,340	181.48	177.39	0.524	13.29	19.89	Daily
No	Yes	Yes	54,379	192.41	181.15	0.524	20.71	27.39	Daily
Yes	No	No	151,089	241.36	248.97	0.524	9.41	17.64	Every other day
Yes	No	Yes	7,850	225.99	243.39	0.520	7.72	16.15	Every other day
Yes	Yes	No	84,254	235.79	244.01	0.524	15.33	23.58	Every other day
Yes	Yes	Yes	58,181	254.03	253.30	0.526	23.57	31.92	Daily

Spending and coupon usage are over the year before the experiment; their thresholds ($179.49 and $17.09) are medians among customers with positive values. "Individual policy" is the unconstrained CLV-optimal assignment, and its schedule is the average number of emails per day in the cell (1 for daily, 0.5 for every other day); "Cell rule" is the uniform frequency with the higher average CLV in the cell. On a random half of the customers, with the cell rule estimated on the other half, the cell rule attains a CLV of $216.36, the individual policy $231.06, and daily emails for all $213.33.

6.5. Sensitivity Analysis

Baseline retention. The value of retaining a customer depends on how much that customer would buy in the future. To see how the results would change for a firm with a different retention environment, we scale the model-predicted annual retention probability under the status quo (0.376) to values between 0.1 and 0.9, scaling the every-other-day probability proportionally (for a baseline of 0.9, every-other-day retention is $0.9 \times 0.388/0.376 = 0.929$). Table 12 and Figure 3 show that the relative value of every-other-day emails and of both personalized policies rises monotonically with baseline retention: when customers are loyal, each avoided opt-out preserves a long stream of purchases, whereas when most customers are one-time buyers, short-run revenue dominates. At the baseline retention of 0.376, the uniform every-other-day policy is roughly neutral and the personalized policies gain 6.8% and 8.3%; at a baseline of 0.6, the same policies gain 8.7%, 13.8%, and 15.6%. The CLV-optimal policy delivers the largest gain at every level. Near a retention probability of one, the multiplier $1/(1-S)$ magnifies any difference in retention, so the right-hand end of the figure is largely mechanical; the qualitative point is that the relative gains rise monotonically with baseline retention.

Discount rate, opt-out sufficiency, and the model inputs. Table 13 collects the remaining sensitivity results; the unconstrained CLV-optimal policy remains the best-performing policy

Table 12 CLV of Each Policy Relative to the Status Quo, by Baseline Retention

Baseline retention	Every other day	Myopic	CLV-optimal
0.1	0.9811	1.0534	1.0648
0.2	0.9869	1.0571	1.0698
0.3	0.9946	1.0622	1.0764
0.376 (Baseline)	1.0025	1.0677	1.0833
0.4	1.0056	1.0700	1.0860
0.5	1.0242	1.0843	1.1021
0.6	1.0868	1.1382	1.1562
0.7	1.2385	1.2760	1.2952
0.8	1.3712	1.3915	1.4194
0.9	1.5010	1.4928	1.5423

Each entry is average CLV under the policy divided by average CLV under daily emails for all, when the baseline annual retention probability (0.376) is rescaled to the value in the first column and the every-other-day probability is rescaled proportionally.

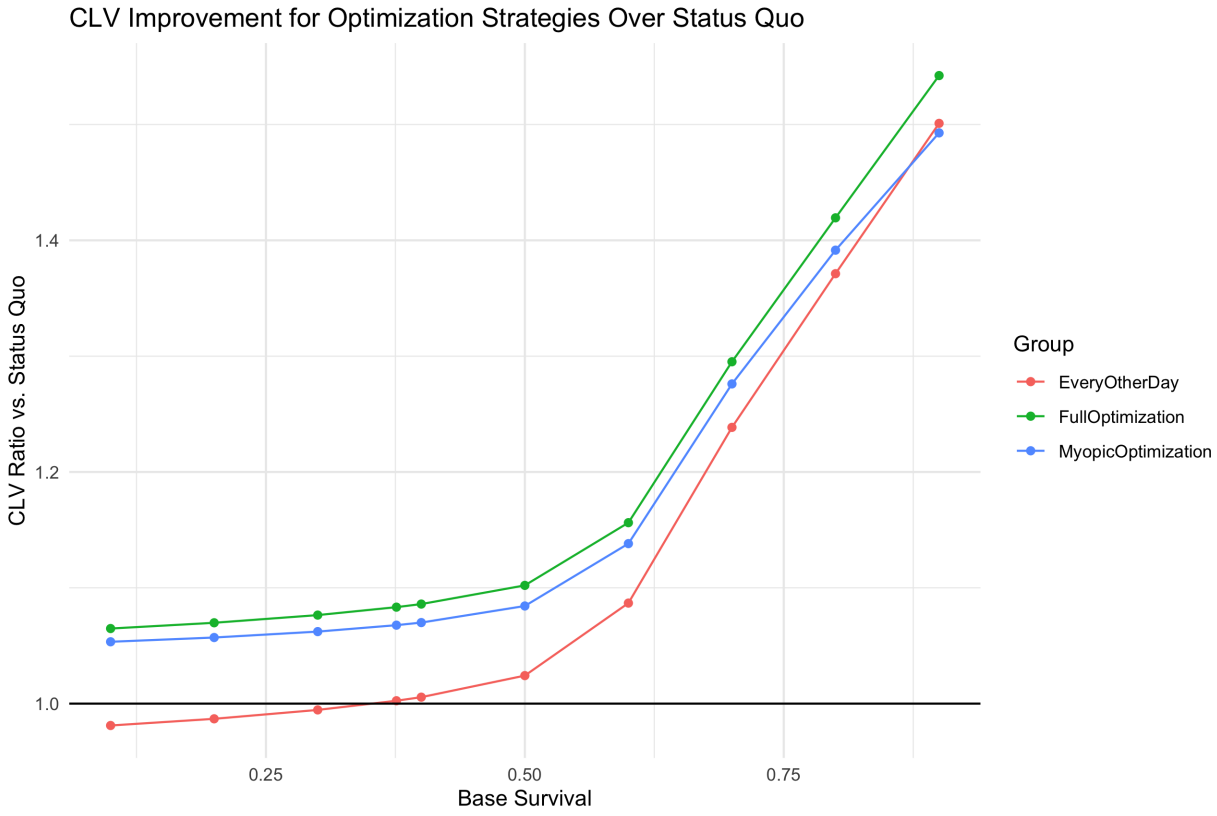


Figure 3 CLV of each policy relative to the status quo as baseline annual retention is varied from 0.1 to 0.9 (baseline value 0.376), with every-other-day retention rescaled proportionally.

throughout, although the uniform policy's ranking relative to the status quo can change. Reoptimizing at annual discount rates up to 10% lowers every CLV level (the status quo falls from \$213.03 to \$182.64) but leaves the percentage gains nearly unchanged (the CLV-optimal gain is

8.0% at $r = 0.10$ vs. 8.3% at $r = 0$, and the constrained gain 4.1% vs. 4.5%), and at most 0.01% of customers change assignment.

Reducing the retention of daily-email customers who remain subscribed by δ, to represent fatigue without opt-out, raises the gains from reduced frequency: the CLV-optimal gain is 8.9%, 9.5%, and 11.2% at $\delta = 0.01$, 0.02, and 0.05, and the constrained gain 5.0%, 5.5%, and 7.1%.

Scaling the post-opt-out revenue loss, the retention probabilities, or both by factors between 0.5 and 1.5, spanning the estimate range in Section 5, moves the uniform every-other-day gain between −3.9% and +7.7%, while the CLV-optimal gain stays between 4.9% and 14.7% and the constrained gain between 1.7% and 11.3%. Resampling the customers in the two historical estimation samples (200 draws) gives 95% intervals of 5.9–7.6% for the myopic gain and 7.0–9.6% for the CLV-optimal gain. Only the uniform policy's CLV gain changes sign across these inputs; across all variations the CLV-optimal policy's average schedule stays between 0.52 and 0.55 emails per day, so most customers are assigned every-other-day emails throughout.

Table 13 Sensitivity of the Policy Results to the Discount Rate, Opt-Out Sufficiency, and the Model Inputs

	Status quo	CLV gain over status quo (%)				Schedule
Case	CLV ($)	Every other day	Myopic	CLV-optimal	CLV-optimal, constrained	(emails/day)
Baseline ($r = 0$, $\delta = 0$)	213.03	0.25	6.77	8.33	4.46	0.525
Annual discount rate r (policies reoptimized at each rate)						
$r = 0.01$	209.53	0.20	6.74	8.29	4.42	0.525
$r = 0.02$	206.15	0.16	6.71	8.25	4.39	0.525
$r = 0.05$	196.64	0.04	6.62	8.15	4.28	0.525
$r = 0.10$	182.64	−0.14	6.50	8.00	4.12	0.525
Retention of daily-email customers who remain subscribed reduced by δ						
$\delta = 0.01$	211.73	0.87	7.25	8.90	4.99	0.524
$\delta = 0.02$	210.44	1.48	7.72	9.47	5.52	0.524
$\delta = 0.05$	206.68	3.33	9.12	11.18	7.14	0.524
Post-opt-out revenue loss scaled by k						
$k = 0.5$	221.85	−2.24	5.57	6.29	2.90	0.534
$k = 0.75$	217.44	−1.02	6.16	7.28	3.60	0.525
$k = 1.25$	208.62	1.57	7.41	9.42	5.47	0.524
$k = 1.5$	204.21	2.95	8.08	10.56	6.60	0.524
Retention probabilities scaled by k						
$k = 0.5$	160.96	−1.39	5.66	6.91	3.04	0.525
$k = 0.75$	183.01	−0.70	6.11	7.51	3.62	0.525
$k = 1.25$	257.80	1.72	7.88	9.60	5.84	0.524
$k = 1.5$	349.05	5.17	10.71	12.56	9.21	0.524
Both historical inputs scaled by k						
$k = 0.5$	167.77	−3.91	4.49	4.91	1.71	0.553
$k = 0.75$	186.85	−1.98	5.51	6.46	2.80	0.526
$k = 1.25$	252.57	3.02	8.53	10.69	6.85	0.524
$k = 1.5$	335.80	7.69	12.06	14.67	11.29	0.524
Bootstrap of the historical inputs (200 draws, 2.5th–97.5th percentiles)						
Retention and post-opt-out revenue inputs	206.5–221.0	−1.36–1.81	5.93–7.61	6.99–9.64	–	0.524–0.525

Schedule is the average number of emails per day under the unconstrained CLV-optimal policy (1 for daily, 0.5 for every other day). The constrained policy re-tunes the budget multiplier in each case so that coupon usage equals its status-quo level. δ is subtracted from the retention probability of subscribed customers under daily emails. k multiplies the proportional revenue decline after opt-out, the retention probabilities, or both. The bootstrap resamples the customers of the two historical estimation samples.

6.6. Managerial Implications

Contact-frequency decisions should account for both immediate sales and expected future customer value. In our experiment, evaluating the two schedules based only on revenue during the experimental week would favor daily emails. Once opt-outs and their downstream consequences are incorporated, however, the unconstrained model favors every-other-day emails for 95% of customers. Campaign evaluation should therefore consider opt-out rates and retention-adjusted customer value alongside short-run conversion and revenue metrics. A rising opt-out rate may signal that contact frequency or promotional content should be adjusted.

The appropriate email frequency also depends on recent customer activity. The model predicts that, for recent purchasers, the short-run revenue benefit of daily reminders outweighs their additional opt-out cost, whereas for other customers the modeled retention benefit favors every-other-day emails. When coupon redemptions are unconstrained, a simple rule that keeps daily emails for recently active customers and moves the remainder to an every-other-day schedule captures nearly all of the gains from customer-level optimization. Importantly, this rule relies on recent purchase activity, an input that is readily available to most retailers. By contrast, segmentation based only on tenure, past-year spending, or coupon usage captures a much smaller share of the gain because these variables do not adequately identify the recent purchasers for whom daily reminders are most valuable.

Accounting for retention becomes especially important when coupon usage is constrained. With coupon redemptions capped at the status-quo level, the myopic policy increases CLV by 2.3%, while the CLV-optimal policy increases it by 4.5%. The resulting assignments are also less uniform than in the unconstrained case. For a budget-constrained retailer, roughly half of the gain therefore comes from accounting for retention rather than from personalization alone.

The value of reducing contact frequency also depends on the underlying retention environment. For loyal, repeat customers, avoiding an opt-out preserves a longer stream of future purchases, making lower contact frequency more valuable. In higher-churn settings, short-run revenue considerations play a larger role and the same intervention generates smaller gains. The same logic can extend to other direct-contact channels, such as push notifications and SMS, where an opt-out or app uninstall may represent both the loss of a communication channel and a signal of customer annoyance. The relevant frequency-response inputs could be estimated using a similar experimental design.

7. Conclusion

The growth of e-commerce and digital marketing has made it inexpensive for firms to reach customers frequently and at scale. This creates a basic tension: more frequent messages can raise

short-run spending but also prompt customers to opt out and disengage over time. Using a randomized field experiment at a large retailer together with historical customer data, we measure this engagement–annoyance trade-off. Moving customers from daily to every-other-day coupon emails reduces opt-outs by 59% while lowering purchase incidence and revenue during the experimental week by 5–8%. Opting out is also followed by declines in spending and the probability of remaining an active customer.

Our findings suggest that email frequency should be treated as a core operating decision, with customer lifetime value guiding the choice of schedule. Firms should track opt-out rates alongside campaign revenue and account for differences in customers' responses to contact. Combining our estimates in a CLV framework yields a simple policy: send daily emails to recently active customers and every-other-day emails to the rest. When coupon redemptions are unconstrained, this rule captures nearly all of the predicted gains from customer-level personalization, raising average lifetime value by 8.3%, and when coupon redemptions are capped at their current level, customer-level optimization yields a predicted gain of 4.5%. Evaluating communication policies on both immediate revenue and subsequent customer behavior can therefore lead to different frequency choices than short-run campaign metrics alone.

Taken together, our results highlight a central managerial takeaway: promotional messages are not free simply because sending them is inexpensive. Accounting for the value of continued customer engagement can help firms design communication policies that perform better over time, and retention-aware, personalized policies can outperform one-size-fits-all approaches.

7.1. Limitations and Directions for Future Work

Our study suggests several directions for further work. The one-week experiment identifies responses to a temporary reduction in email frequency among customers otherwise receiving daily emails. Longer experiments would help establish whether these responses attenuate or accumulate under a sustained policy. The post-experiment estimates and sensitivity analyses help assess the projections, while sustained responses remain to be measured directly. A multi-arm design could also extend the comparison beyond the two email schedules studied here.

The longer-run consequences of opt-out are estimated using observational data from customers who made at least one purchase in 2024 and are transferred, conditional on tenure, to the experimental population. The predicted gains from personalization remain positive across the range of historical estimates considered. Following the randomized customers over a longer horizon would provide more direct evidence on the longer-run effects of email frequency and help assess these policy projections.

The parsimonious CLV model also offers opportunities for refinement. As discussed in Section 6.1, allowing opt-out status to persist across years, allowing post-opt-out revenue responses to vary

across customers, and updating customer covariates would provide a richer account of customer behavior. The recent-activity policy could then be evaluated as a rolling rule; the static model may overstate the value of maintaining daily emails for recent purchasers. Customer-level assignments also depend on predictions of revenue, opt-out, and coupon usage from one-week linear interaction models, and the budget constraint applies to predicted coupon redemptions. Longer observation periods would support evaluating and refining these predictions.

Finally, our setting involves a single retailer with relatively infrequent, discretionary purchases and a broadcast coupon program. The underlying trade-off between short-run engagement and longer-run customer disengagement may extend to other direct-marketing settings, but its magnitude is likely to depend on factors such as baseline retention, purchase frequency, and offer design. Our baseline-retention analysis in Section 6.5 illustrates how the value of reducing contact frequency can vary with the customer relationship and provides a starting point for such comparisons.

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Appendix A: Robustness Checks: Alternative Functional Forms

Table 14 re-estimates the experimental treatment effects using logistic regressions for opt-out and purchase incidence and a Poisson regression for item quantity, with date fixed effects. All three effects remain negative and statistically significant. The opt-out coefficient of -0.896 corresponds to an odds ratio of $e^{-0.896} = 0.41$, consistent with the reduction reported in Table 4. These coefficients are on the log-odds or log-mean scale and are not directly comparable with the linear-model coefficients.

Table 15 re-estimates Equation (1) with log-transformed revenue, item quantity, and price per item. Fewer emails lower short-term revenue and item quantity, with no statistically significant change in price per item. Table 16 reports the OLS and IV estimates of Section 5.2 with log revenue, and Table 17 re-estimates the retention regressions of Table 7 by logistic regression. Conclusions are unchanged.

Table 14 Treatment Effects of Reducing Email Frequency, Nonlinear Models

	P(Unsubscribe)	Item Quantity	P(Purchase)
Treated	-0.896^{***}	-0.105^{***}	-0.050^{**}
	(0.069)	(0.015)	(0.020)
Observations	4,222,071	4,222,071	4,222,071
Fixed Effects	Date	Date	Date
Model	Logistic	Poisson	Logistic

Note: *p<0.1; **p<0.05; ***p<0.01
Estimates use the experimental week. Logistic and Poisson coefficients are on the log-odds and log-mean scales, respectively. Revenue and price per item are not re-estimated.

Table 15 Treatment Effects of Reducing Email Frequency, Log-Transformed Outcomes

	Dependent variable:		
	log(1+Revenue)	log(1+Item Quantity)	log(Price Per Item)
	(1)	(2)	(3)
Treated	-0.0006^{***}	-0.00015^{***}	-0.010
	(0.0002)	(0.00005)	(0.019)
Observations	4,222,071	4,222,071	9,635
Fixed Effects	Date	Date	Date

Note: *p<0.1; **p<0.05; ***p<0.01

Table 16 Revenue in the Month After Opt-Out, Log-Transformed Revenue

	Dependent variable:	
	log(1 + Month Revenue)	
	(1)	(2)
Previous Month Revenue	0.00023***	
	(0.00001)	
Previous Month Unsubscribed	−0.174***	−0.349***
	(0.008)	(0.075)
Observations	1,095,832	1,016,252
Fixed Effects	Month	Customer, month
Estimator	OLS	IV

Note: *p<0.1; **p<0.05; ***p<0.01
Column (2) instruments opt-out with the change in email opens, as in Table 6.

Table 17 Retention After Opt-Out, Logistic Regression

	Dependent variable:	
	Survival	
	After Month 1	After Months 1–6
	(1)	(2)
Unsubscribed	−1.157***	−1.002***
	(0.275)	(0.063)
Tenure (years)	0.057***	0.052***
	(0.003)	(0.002)
Unsubscribed×Tenure	−0.015	0.002
	(0.048)	(0.008)
Constant	−1.474***	−2.363***
	(0.023)	(0.014)
Observations	23,465	118,941
Model	Logistic	Logistic

Note: *p<0.1; **p<0.05; ***p<0.01
Samples and windows as in Table 7.

Appendix B: Heterogeneous Treatment Effects: Median-Split Moderators

Tables 18–21 re-estimate Equation (2) with median-split moderators. The email-opens interaction in the opt-out equation and the item-quantity interaction in the revenue equation retain their signs and significance; the tenure interaction in the opt-out equation is significant only at the 10% level in the combined model and not on its own. Table 22 re-estimates the heterogeneity in opt-out's consequences (Section 5.4) with binary moderators; none of the opt-out interactions is significant.

Table 18 HTEs: Customer Tenure, Median Split

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Tenure>2,577 Days	−0.00518**	0.025
	(0.00212)	(0.025)
Treated	−0.02271***	−0.050**
	(0.00216)	(0.025)
Tenure>2,577 Days×Treated	0.00472	0.027
	(0.00305)	(0.036)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date

Note: *p<0.1; **p<0.05; ***p<0.01

The tenure threshold is the median on October 25, 2024.

Table 19 HTEs: Item Quantity in Past 7 Days, Median Split

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Item Quantity>0	0.02864***	1.807***
	(0.00986)	(0.117)
Treated	−0.02035***	−0.022
	(0.00153)	(0.018)
Item Quantity>0×Treated	−0.00229	−1.250***
	(0.01402)	(0.166)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date

Note: *p<0.1; **p<0.05; ***p<0.01

Table 20 HTEs: Email Opens in Past 7 Days, Median Split

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Email Opens>0	0.03513***	0.025
	(0.00213)	(0.025)
Treated	−0.00633***	−0.031
	(0.00220)	(0.026)
Email Opens>0×Treated	−0.02798***	−0.012
	(0.00305)	(0.036)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date

Note: *p<0.1; **p<0.05; ***p<0.01

Table 21 HTEs: All Variables Combined, Median Split

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Tenure>2,577 Days	−0.00605***	0.024
	(0.00213)	(0.025)
Item Quantity>0	0.02740***	1.806***
	(0.00986)	(0.117)
Email Opens>0	0.03523***	0.022
	(0.00213)	(0.025)
Treated	−0.00899***	−0.031
	(0.00267)	(0.032)
Tenure>2,577 Days×Treated	0.00547*	0.027
	(0.00305)	(0.036)
Item Quantity>0×Treated	−0.00126	−1.250***
	(0.01402)	(0.166)
Email Opens>0×Treated	−0.02812***	−0.010
	(0.00305)	(0.036)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 22 Heterogeneity in Revenue After Opt-Out, Binary Moderators

	Dependent variable:		
	Month Revenue ($)		
	(1)	(2)	(3)
Previous Month Revenue	0.087***	0.086***	0.112***
	(0.001)	(0.001)	(0.001)
HTE	5.549***	2.869***	−27.404***
	(0.243)	(0.253)	(0.391)
Previous Month Unsubscribed	−9.508***	−10.266***	−7.664***
	(1.003)	(2.238)	(0.854)
Previous Month Unsubscribed×HTE	−0.075	−0.332	−0.263
	(1.496)	(2.373)	(1.736)
Observations	1,095,832	1,095,832	1,095,832
Fixed Effects	Month	Month	Month
HTE	Tenure	Email Opens	Item Quantity

Note: *p<0.1; **p<0.05; ***p<0.01

Specification of Table 6, column (1). HTE is an indicator for tenure above its sample median (1,247 days), positive email opens, or positive item quantity.

Appendix C: Heterogeneous Treatment Effects: Single-Moderator Models and the Consequences of Opt-Out

Tables 23–25 report single-moderator versions of Equation (2) underlying the combined specification in Table 5. Table 26 interacts the opt-out indicator in the specification of Table 6, column (1), with each moderator; the item-quantity interaction is negative and significant, while the tenure and email-opens interactions are not.

Table 23 HTEs: Customer Tenure

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Tenure (years)	−0.00080***	0.0064**
	(0.00023)	(0.0028)
Treated	−0.02566***	−0.077**
	(0.00302)	(0.036)
Tenure×Treated	0.00069**	0.0052
	(0.00034)	(0.0040)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 24 HTEs: Item Quantity in Past 7 Days

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Item Quantity	0.00313	6.632***
	(0.00249)	(0.029)
Treated	−0.02040***	0.093***
	(0.00153)	(0.018)
Item Quantity×Treated	0.00185	−6.189***
	(0.00485)	(0.057)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 25 HTEs: Email Opens in Past 7 Days

	Dependent variable:	
	P(Unsubscribe)×100	Revenue ($/Day)
	(1)	(2)
Email Opens	0.01116***	−0.005
	(0.00083)	(0.010)
Treated	−0.01185***	−0.062***
	(0.00189)	(0.022)
Email Opens×Treated	−0.00933***	0.026*
	(0.00119)	(0.014)
Observations	4,222,071	4,222,071
Fixed Effects	Date	Date
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 26 Heterogeneity in Revenue After Opt-Out

	Dependent variable:		
	Month Revenue ($)		
	(1)	(2)	(3)
Previous Month Revenue	0.087***	0.087***	0.060***
	(0.001)	(0.001)	(0.001)
HTE	0.518***	0.052***	7.752***
	(0.023)	(0.004)	(0.147)
Previous Month Unsubscribed	−9.140***	−9.989***	−7.017***
	(0.968)	(0.935)	(0.796)
Previous Month Unsubscribed×HTE	−0.087	0.030	−8.066***
	(0.146)	(0.032)	(0.683)
Observations	1,095,832	1,095,832	1,095,832
Fixed Effects	Month	Month	Month
HTE	Tenure (years)	Email Opens	Item Quantity
Note:			*p<0.1; **p<0.05; ***p<0.01

Specification of Table 6, column (1). Tenure is measured on January 1, 2024; email opens and item quantity are measured in the opt-out month.

Appendix D: Robustness Checks: Instrumental Variables

Opt-out's negative revenue effect is similar with a logit first stage (Table 27) and with a leave-one-out instrument based on other customers' mean change in opens (Table 28). Both IV estimates include customer and month fixed effects. Table 29 reports the first stage for the IV estimate in Table 6, column (2).

Table 27 IV with a Logit First Stage

	Dependent variable:
	Month Revenue ($)
Previous Month Unsubscribed	−9.298***
	(2.442)
Observations	1,016,252
Fixed Effects	Customer, month
Instrument	Logit fitted probability
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 28 IV with a Leave-One-Out Instrument

	Dependent variable:
	Month Revenue ($)
Previous Month Unsubscribed	−8.771***
	(2.976)
Observations	1,016,252
Fixed Effects	Customer, month
Instrument	Other customers' mean change in opens
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 29 IV First Stage

	Dependent variable:
	Previous Month Unsubscribed
Change in Email Opens	−0.001151***
	(0.000008)
Observations	1,016,252
Fixed Effects	Customer, month
Partial F	20,225.0
Note:	*p<0.1; **p<0.05; ***p<0.01

Appendix E: Additional Historical-Panel Specifications

Table 30 compares opt-out estimates for the 1,095,832 customer-months of Section 5.2. Column (1) repeats Table 6, column (1). Column (2) replaces lagged revenue with customer fixed effects. Column (3) adds lagged revenue to that model and illustrates the short-panel bias discussed by Nickell (1981): its lagged-revenue coefficient is −0.029, compared with +0.086 in column (1).

Coefficient stability. Following Oster (2019), we track the opt-out coefficient as controls are added: without controls it is −7.71 ($R^2 = 0.0001$), with month effects −8.03 ($R^2 = 0.0021$), and with month effects and lagged revenue −9.85 ($R^2 = 0.0167$). Because the controls move the coefficient away from zero, the bias-adjusted coefficient for $\delta = 1$ is −10.50 with $R_{\max} = 1.3\tilde{R}$ and −12.01 with $R_{\max} = 2\tilde{R}$, where $\tilde{R} = 0.0167$ is the R^2 of the full specification.

Cohort survival analysis. Table 31 reports, for each opt-out month $m = 1, \ldots, 6$, survival over months $m+1$ to $m+6$ among customers purchasing in month m who were subscribed at its start, by permanent opt-out in month m, and the opt-out coefficient from the specification of Table 7 estimated within the cohort.

Table 30 Revenue in the Month After Opt-Out: Alternative Estimators

	Dependent variable:		
	Month Revenue ($)		
	(1)	(2)	(3)
Previous Month Unsubscribed	−9.853***	−5.831***	−7.935***
	(0.745)	(1.008)	(1.008)
Previous Month Revenue	0.086***		−0.029***
	(0.001)		(0.001)
Observations	1,095,832	1,095,832	1,095,832
Fixed Effects	Month	Customer, month	Customer, month
Note:			*p<0.1; **p<0.05; ***p<0.01

Table 31 Six-Month Survival by Opt-Out Cohort

Opt-out month m	Customers	Permanent opt-outs	Survival, opt-outs	Survival, others	Opt-out coefficient	(s.e.)
1	23,465	326	0.064	0.188	−0.090	(0.027)
2	23,637	506	0.067	0.186	−0.076	(0.022)
3	24,090	809	0.084	0.173	−0.073	(0.017)
4	22,711	846	0.095	0.169	−0.061	(0.017)
5	20,908	836	0.063	0.172	−0.094	(0.016)
6	16,899	679	0.068	0.174	−0.072	(0.018)
Pooled, cohort fixed effects	131,710	4,002			−0.076	(0.008)

Survival indicates at least one purchase in months $m+1$ to $m+6$. The coefficient is on the permanent opt-out indicator in a regression that also includes tenure and its interaction with opt-out. A test of equal opt-out coefficients across cohorts gives $F = 1.27$ ($p = 0.28$).

Appendix F: Additional Policy-Model Results

Table 32 compares predicted outcomes under daily and every-other-day emails within median splits of tenure, past-year spending, and past-year coupon usage. These comparisons summarize group-level predictions; they do not determine the assignments of the optimized policy, which uses each customer's joint covariates (Section 6.4). For longer-tenured customers, every-other-day emails yield the higher average CLV ($248.30 vs. $241.80): their opt-out response to frequency is smaller, but their higher baseline retention makes each retained customer more valuable. For shorter-tenured customers the averages favor daily emails ($184.25 vs. $178.90). Daily emails also have the higher average CLV for high spenders and high coupon users, whereas every-other-day emails yield the higher average CLV for low spenders and low coupon users. As Table 11 shows, these group averages are driven by the few recently active customers within each group.

Table 32 Predicted Outcomes Under Uniform Policies by Customer Segment

Email Policy	CLV ($)	Revenue Rate ($/Month)	E[Revenue] ($/Year)	Retention (/Year)	Coupon Usage ($/Year)	Schedule (Emails/Day)
Shorter tenure ($\leq$ 2,583 days)						
Daily (status quo)	184.25	11.45	125.50	0.319	11.08	1
Every other day	178.90	10.27	119.30	0.332	9.38	0.50
Longer tenure ($>$ 2,583 days)						
Daily (status quo)	241.80	12.12	135.10	0.432	13.75	1
Every other day	248.30	11.60	135.08	0.445	13.71	0.50
High spenders (past-year spending $>$ \$179.49)						
Daily (status quo)	214.42	11.94	131.31	0.375	17.41	1
Every other day	213.28	10.94	127.13	0.388	12.48	0.50
Low spenders (past-year spending $\leq$ \$179.49)						
Daily (status quo)	211.79	11.65	129.30	0.376	7.94	1
Every other day	213.82	10.93	127.30	0.388	10.71	0.50
High coupon users (past-year coupon usage $>$ \$17.09)						
Daily (status quo)	221.11	12.35	134.70	0.379	20.26	1
Every other day	217.40	11.11	128.63	0.392	13.22	0.50
Low coupon users (past-year coupon usage $\leq$ \$17.09)						
Daily (status quo)	210.86	11.63	129.10	0.375	10.31	1
Every other day	212.53	10.89	126.80	0.387	11.10	0.50

Columns as in Table 9. Thresholds are the medians used in Table 11.